

Daniel Lee

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EDUCATION

Purdue University

Bachelor of Science in Data Science

West Lafayette, IN

August 2022 – May 2026

RESEARCH FOCUS

Focused on solving real-world data challenges that result in reliable, actionable insight with special interests in cloud infrastructure and human-in-the-loop designs. My experience is centered on large-scale data, NLP, and distributed computing, as well as designing pipelines and analytical models that act as decision-support tools.

Relevant Coursework: Data Structures and Algorithms, Data Mining and Machine Learning, Large Scale Data Analytics

PUBLICATIONS AND PATENTS

Undergraduate Student Research

The Data Mine (Purdue University)

August 2022 – May 2023

West Lafayette, IN

- Participated in a **joint research** program with the CDC Protect initiative to identify factors for pediatric melatonin overdose and identify impact of misleading product labels
- Analyzed over **347** products from the **NIH dietary supplement database** to establish a reference schema and linguistic baseline for product labels, providing a benchmark for comparison with online products
- Built **NLP** pipelines with **HuggingFace transformers** to perform topic modeling and sentiment analysis on **3,500+** Amazon reviews and **410+** Reddit posts, identifying recurring themes in melatonin product discussion

PROJECTS

CampusBuzz - Node.js, Next.js, PostgreSQL

August 2025 – January 2026

- Addressed the absence of real-time student feedback at Purdue career fairs by architecting a full-stack platform that enables live ratings, reviews, and company analytics for comprehensive insight into career fairs and companies
- Designed and implemented a relational **PostgreSQL** schema supporting multi-role users (students, recruiters, admins), career fairs, companies, reviews, and engagement metrics over large volumes of user-generated data
- Integrated frontend with **Next.js**, **React**, and **Tailwind CSS** for interactive menus and polished UI/UX
- Developed analytics pipelines that aggregated review data into actionable insights such as average ratings, sentiment trends, and company comparisons, allowing students to prioritize booths and recruiters to evaluate engagement

WORK EXPERIENCE

Software Engineer Intern – Node.js, Azure

Daedalus Labs

July 2024 – October 2024

San Francisco, CA

- Faced with fragmented blockchain webhook payloads that prevented reliable transaction tracking, implemented a **Node.js** ingestion pipeline on **Microsoft Azure** to transform raw event streams into structured **JSON** records and log thousands of Bitcoin Rune transactions into **Azure SQL** for reliable retrieval and downstream analysis
- Engineered an automated monitoring and alerting system that detected Bitcoin and Fractal transaction events and pushed live notifications via Discord integrations and a browser-based Chrome extension
- Conducted exploratory analysis of the emerging Fractal transaction protocol to preemptively identify integration strategies, presenting findings in Agile sprint reviews to guide early-stage monitoring

Data Analyst Intern – Python, Docker, Parquet

NSF I-GUIDE

August 2023 – May 2024

West Lafayette, IN

- Analyzed flood-inundation raster data from **345+** dams in conjunction with census-block socioeconomic indicators to characterize population-level vulnerability across affected regions
- Re-engineered core pipelines into distributed **Apache Spark** and **Sedona** jobs, enabling large-area spatial joins on HPC infrastructure
- Diagnosed computational bottlenecks in raster-heavy workloads using Python profiling tools, guiding targeted optimizations in data loading and spatial operations
- Converted raw **GeoTIFF** archives into parallel-optimized **Parquet** and **Zarr** formats to improve downstream I/O performance

TECHNICAL SKILLS

Languages: Java, Python, Javascript, C++, R

Tools/Frameworks: Flask, React, Node.js, Git, Jupyter Notebook, Docker, Supabase, Selenium, Azure, PostgreSQL

Spoken Languages: English (Native), Korean (Native), Spanish (Intermediate)